

Europe needs a strategy to close the artificial intelligence compute gap

In its bid to compete with the US on AI, Europe could learn from both China and from the classic Airbus industrial policy case

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The race to dominate artificial intelligence is to a large extent a race for compute, or processing power. Advanced AI systems require vast quantities of specialised chips able to run the calculations that train and operate large models. Whoever controls that infrastructure shapes the technology and sets the economic and strategic terms on which others access it.

That race is between the United States, which dominates frontier chip design through firms such as NVIDIA, and China, which is spending enormous political capital and resources to reduce its dependence on that same ecosystem. Both understand that hardware forms a chokepoint. The United States uses export controls to limit Chinese access to chips (CRS, 2025). China has responded with diplomatic pressure, regulatory capture of its domestic market and a deliberate domestic industrial build-out, while Chinese companies have found ways around export controls (Juniewicz, 2026).

The outcome of the US/China contest could shape the structure of the global AI economy for decades. Meanwhile, for European firms, the AI hardware stack – chips, software interfaces, data centres – will increasingly determine the conditions under which they develop AI applications: at what cost, subject to what regulation and on whose terms. If Europe remains dependent on either US or Chinese compute infrastructure, it faces a structural loss of economic autonomy as AI becomes the general-purpose technology of the century.

Europe, though comparable in market size to the US and China, does have upstream hardware assets – a global monopoly on extreme ultraviolet lithography equipment, the single most critical tool in the production of advanced semiconductors, in the Dutch firm ASML, for example. Belgium's IMEC leads research into the tiniest semiconductors. Yet Europe exports these advantages to the US, South Korea and Taiwan. The question is whether this is a permanent condition or a policy choice, and if so, whether there is an alternative.

China: diplomacy and domestic build-out

China is closing in rapidly on the US in terms of hardware gap through a deliberate two-track strategy. For Europe, this should be instructive: unlike the US, which has been at the technology frontier throughout, China started from a position of dependency on foreign hardware. Europe needs to replicate China's trajectory in terms of closing the hardware gap.

The first track in China's strategy is diplomatic. Since 2022, US export controls have systematically restricted China's access to advanced semiconductors (CRS, 2025), but China has used its leverage over rare earths and market access to extract concessions. Under President Donald Trump, a more transactional approach has been taken, with the lifting of some restrictions on exports of advanced AI hardware to China.

In particular, in January 2026, the US Department of Commerce shifted its review policy for NVIDIA H200 chips destined for China from a presumption of denial to case-by-case approval, subject to a 25 percent tariff. Orders from Chinese tech companies in 2026 could reach \$14 billion¹. Trump's White House has downplayed controls while approving higher-tier chip exports and suspending further restrictions². China has also circumvented controls by training AI models in Southeast Asia and Europe³.

The second track is domestic. Huawei remains the central player. Chinese firms unable to source NVIDIA hardware use its Ascend chips, which power 41 percent of China's data centres⁴. Alibaba's chip unit T-Head is second in domestic shipments, followed by Baidu.

Meanwhile, China's government directs state-owned enterprises to prioritise domestic hardware, creating a captive market that generates revenues large enough to fund the

next round of R&D. Huawei's AI chip revenue could be \$12 billion in 2026, up from \$7.5 billion in 2025⁵, revenue to fund the improvements geared towards closing the performance gap.

The performance gap is increasingly irrelevant

Progress notwithstanding, Huawei's Ascend 910C delivers only approximately 60 percent of NVIDIA's H100 inference performance and Chinese firms have been reluctant to switch: many top AI models, including DeepSeek's V3, are still trained on NVIDIA hardware. Industry has converged on NVIDIA's software interface – CUDA – and switching to Huawei's CANN platform would require rewriting the code of AI models, potentially resulting in a performance drop (Ottinger and McMahon, 2025). DeepSeek spread rapidly because it is built on CUDA, for example. CANN does not have CUDA's massive documentation library and English language community forums.

Nevertheless, Huawei aims to double the capabilities of its chips by the end of 2027⁶. China is buying time by pushing for the most advanced chips from the US while supporting deployment of Chinese chips through subsidies, captive demand and below-market energy prices for AI firms using domestic chips. The reluctance of Chinese AI companies to abandon CUDA may be a transitional friction as Huawei's stack improves and its CANN platform matures, perhaps over the next three to five years.

Europe's missing scale

On the model side, France's Mistral represents the European Union's most serious attempt at a competitive AI presence, but it operates at a scale that barely registers against frontier US or Chinese labs (HAI, 2026, table 1.1.1). Europe's constraint is not primarily talent or ideas, but access to compute and capital – Mistral trained its flagship models on Microsoft Azure's supercomputing infrastructure⁷ and opened a Palo Alto office specifically to attract engineers and AI scientists and to access Silicon Valley venture capital⁸. Mistral Compute, a European AI infrastructure platform, has been built in partnership with NVIDIA. Intended to offer an alternative to US-based cloud providers it is relatively tiny, with €830 million raised to build data centres in France and Sweden⁹.

Unlike China, the EU has no coordinated mechanism to direct public procurement towards European AI hardware or to generate the revenues that fund R&D. EU demand

is fragmented across twenty-seven member states, each acting largely alone. The EU is planning to improve public procurement for sovereign cloud and AI infrastructure, with a 'European preference' for public-sector purchases of critical technologies¹⁰, but unlike in China, EU policymakers have limited leverage over private-sector demand and limited tools generate the captive market that powers Huawei's improvement cycle.

There is no realistic prospect for Europe to switch to its own AI chip production in the short to medium term. The EU could, however, learn from China's two-track strategy: continue to buy AI chips from NVIDIA and other US suppliers while investing in its own advanced chips industry. The building blocks already present in the EU for such an industry need to be orchestrated into a coherent industrial strategy for domestic production.

AI and Airbus

In European industrial policy debates, Airbus is typically cited as evidence that coordinated EU action can produce a globally competitive champion when no individual member state can¹¹. But the analogy is often applied without any account of what actually made Airbus work. In relation to AI hardware, it is clear that Europe should avoid a distributionally inspired negotiation in which member states argue over which component can be built in their territory. In building European advanced chips to serve Europe's own compute capacity, the typical mix of political interest and industrial lobbying should be avoided in favour of genuine technological complementarities based on comparative advantage.

In fact, what defined Airbus was that the founding division of work and investment was not arbitrary. France had Aérospatiale, already offering expertise in fuselage systems and avionics integration. Germany provided manufacturing depth and precision engineering. The United Kingdom contributed wing design, in which British Aerospace had internationally recognised capability. Spain's CASA had experience in the manufacturing of airframes. The political negotiation on where to do what shaped the broad structure, but the substance of each assignment reflected real industrial logic.

Crucially, Airbus evolved from a loose intergovernmental consortium into a proper integrated company, with unified management, shared intellectual property and the ability to make decisions that cut across national preferences. It became competitive not because of the political deal that launched it, but because the deal was eventually subordinated to commercial logic.

A European AI hardware version of this model would need to start from the technology stack and work outward to the institutional structure, not the other way around. In assembling an EU AI consortium, the first question should not be ‘which country gets a piece of this?’ but ‘where do European firms have genuine comparative advantage in the AI hardware value chain, and what does a competitive stack require?’ Clear existing capabilities are ASML’s position in extreme ultraviolet lithography and IMEC’s role in advanced fabrication research. But Europe also has photonics firms, power semiconductor manufacturers including Infineon and STMicroelectronics, and precision optics companies such as Carl Zeiss. What matters is the comparative advantage of each firm that ends up in the consortium. Selection should be competitive and, as much as possible, determined by the technology rather than the map of European capitals.

The institutional vehicle matters too. The legal and governance structure should enable participating firms to pool intellectual property, share development costs and make investment decisions collectively. It should also give the resulting entity enough autonomy to respond to market signals rather than ministerial preferences.

In this respect, the EU has tools that did not exist when Airbus was conceived. The Important Projects of Common European Interest (IPCEIs) provide a legal framework for joint R&D investment across member states with state aid exemptions¹². The 2023 IPCEI on Microelectronics and Communication Technologies, involving 56 companies across 14 member states, assembled €21.8 billion in combined public and private investment¹³. In the European Chips Act, the EU also has a framework intended to facilitate an expansion of the EU semiconductor industry (Poitiers and Schenk, 2026). In other words, Europe is not starting from scratch.

On the demand side, Europe cannot replicate China’s strategy directly as the EU has limited leverage over private-sector procurement and coercion would be both legally problematic and economically self-defeating. But coordinated public procurement across EU countries for AI compute used in public administration, healthcare, defence and research infrastructure could create an initial demand base substantial enough to fund development. Plans for a common EU marketplace for cloud capacity and a ‘European preference’ in public procurement for critical technologies (see footnote 10) are a start, but these should be linked explicitly and durably to hardware sourcing commitments, not just to cloud-service providers that may themselves depend on non-European chips.

Yet this strategy also brings real costs for European companies, and those costs deserve honest reckoning. European firms that switch from NVIDIA hardware to domestic alternatives in the short to medium term will face a performance penalty: slower training, higher energy consumption per computation and the friction of migrating from the deeply embedded CUDA software ecosystem to less-mature domestic equivalents. Those are genuine competitive disadvantages, not transitional inconveniences.

The question of whether the EU should compensate European companies for these costs will therefore arise. A well-designed compensation mechanism could include R&D subsidies, preferential pricing for public-sector compute contracts, accelerated depreciation allowances for domestic hardware investment or mandated market share thresholds accompanied by price-support schemes. These could make the difference between an effective industrial policy and one that companies simply ignore. But without some form of compensation or protection, the incentive for European companies to absorb short-term losses in exchange for long-term strategic autonomy is weak. It is worth noting that early Airbus subsidies involved explicit compensation for competitive disadvantage during a build-up phase.

None of this is a short-term proposition. Airbus took twenty years to become genuinely competitive with Boeing. A European AI hardware initiative would face a steeper climb in a faster-moving technology environment but the baseline against which Europe should measure itself is not catching up with NVIDIA next year. Rather, it is avoiding a future in which the compute infrastructure that underlies European AI is controlled permanently by either Washington or Beijing, on terms that Europe did not negotiate and cannot revise.

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